Killiivalavan K A

killiivalavan.inbox@gmail.com | Killiivalavan | Portfolio | Killiivalavan

SUMMARY

Python developer experienced in building backend systems and AI-integrated applications including RAG pipelines, agentic automation, and async architecture. Comfortable across Python, Go, and TypeScript, with hands-on experience in self-hosted infrastructure and shipping production tools.

WORK EXPERIENCE

Yodigispace

Junior Programmer — Aug 2025 - Present

- Built and shipped two full modules (HR and Production) for yodsBMS, an AI-integrated Business Management System, integrating Production with existing inventory and stock management modules for manufacturing companies as clients.
- Designed and implemented a RAG (Retrieval-Augmented Generation) pipeline for a Document Management System module, enabling natural-language document querying, currently extending it with GraphRAG to improve retrieval performance.
- Designed and implemented approval based business logic workflows, routing items through role-based reviews and sign-off stages.
- Reduced application server cold start time from 1+ minute to under 10 seconds through backend optimization.
- Worked across Python, databases (SQLite, PostgreSQL), and REST APIs, applied cryptographic techniques for securing variables and API keys handling and authentication flows.

TECHNICAL SKILLS

- Languages: Python, JavaScript, TypeScript, SQL, Golang
- Frameworks: React, Next.js, Tailwind CSS, FastAPI, Flask, Django, pytest, Node.js, Express.js
- Tools: Git, GitHub, Docker, MongoDB, MySQL

PROJECTS

D.A.I.S.Y (Ongoing)

- Built a fully asynchronous agentic voice assistant pipeline targeting sub 2 second end-to-end latency.
- Architected a client-server split over Tailscale, self-hosted on a home Ubuntu server, integrating 13+ tools enabling autonomous background coding sessions (via OpenCode), shell command execution, proactive task/reminder management, file I/O, and web search with text extraction
- Designed a hybrid LLM routing layer - large model for conversational turns, smaller models for RAG, memory consolidation, and tool parsing to balance responsiveness and cost

chesshell (2026)

- Built a CLI chess client in Go rendering a graphical board directly in the terminal
- Integrated the Lichess API for daily puzzles with zero auth-access and a local Stockfish engine for AI opponents
- Implemented real-time multiplayer using Cloudflare Durable Objects with shareable room codes

the-B-Side (Ongoing)

- Designed and built a social platform for logging, rating, and reviewing albums/tracks, with discovery features for finding new music, previously deployed with active users.
- Built the initial data layer on Supabase (auth, database) with a Vite-powered frontend.
- Currently undergoing a full rebuild of both frontend and backend based on learnings from the first version

tabstone (2025)

- Cross-browser extension to archive/restore open tabs with persistent local storage, published and maintained on both Chrome Web Store and Firefox Add-ons

EDUCATION

SRM University - B.tech CSE with specialization in AI and Data Science — 2021 - 2025

CERTIFICATIONS

Artificial Intelligence Analyst - IBM — June 2024

Introduction to Python - IBM — Nov 2023

Cloud Application Developer - IBM — June 2023